

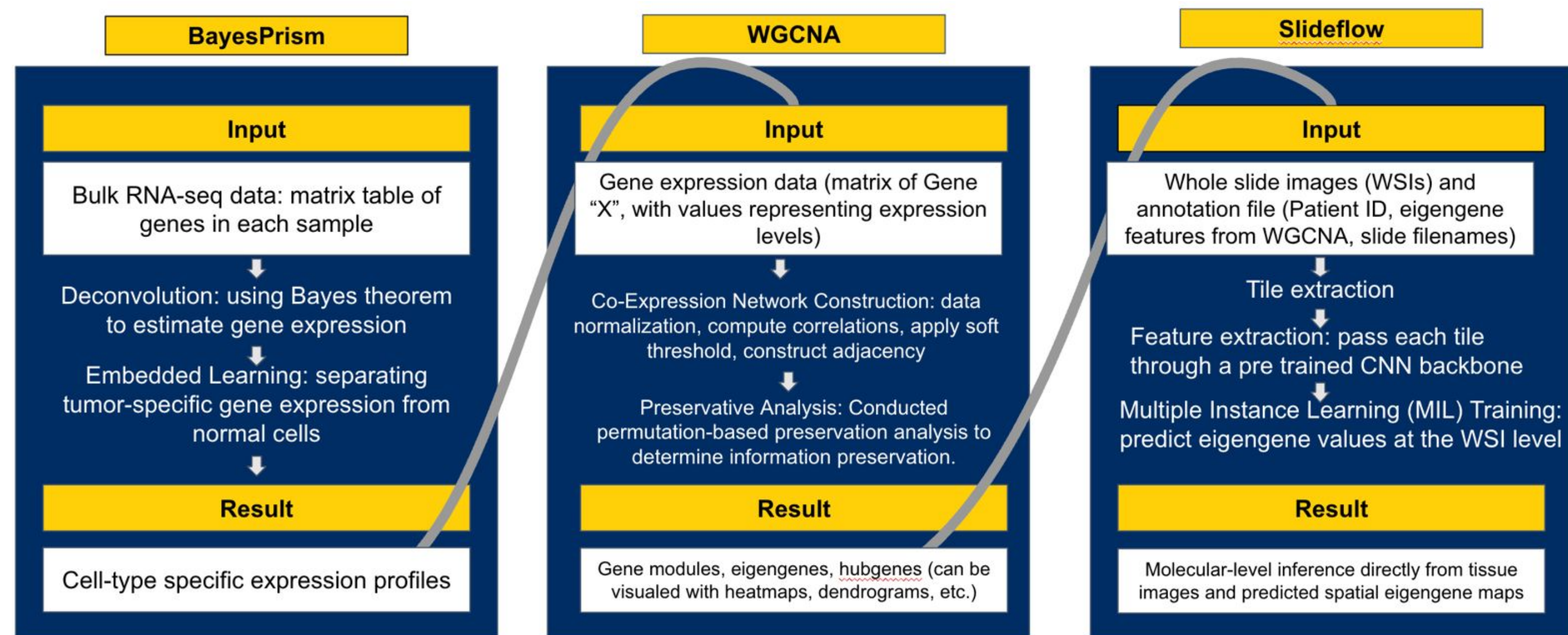
A BayesPrism–WGCNA–Slideflow Pipeline for Inferring Spatial Gene Expression Patterns in Glioblastoma

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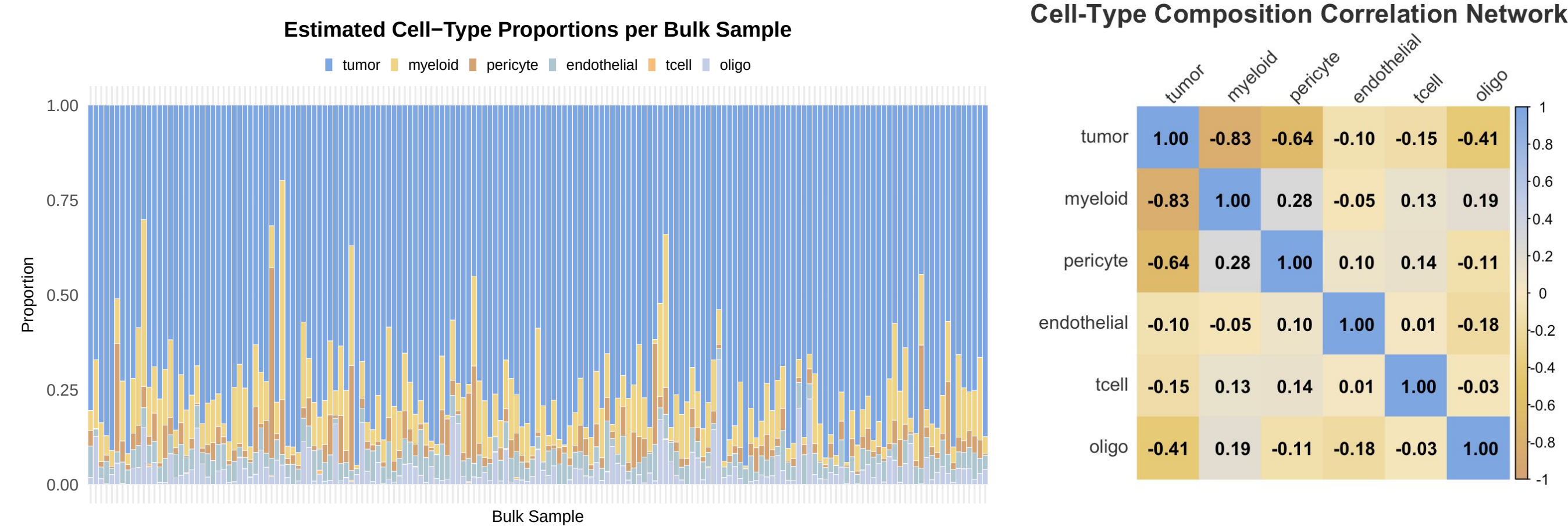


Framework



BayesPrism

Bayesian Deconvolution of Bulk Transcriptomes. We applied **BayesPrism**, a hierarchical Bayesian model integrating single-cell references, to infer latent cell-type proportions (θ) and cell-type-specific expression from bulk RNA-seq, enabling probabilistic dissection of the tumor microenvironment.

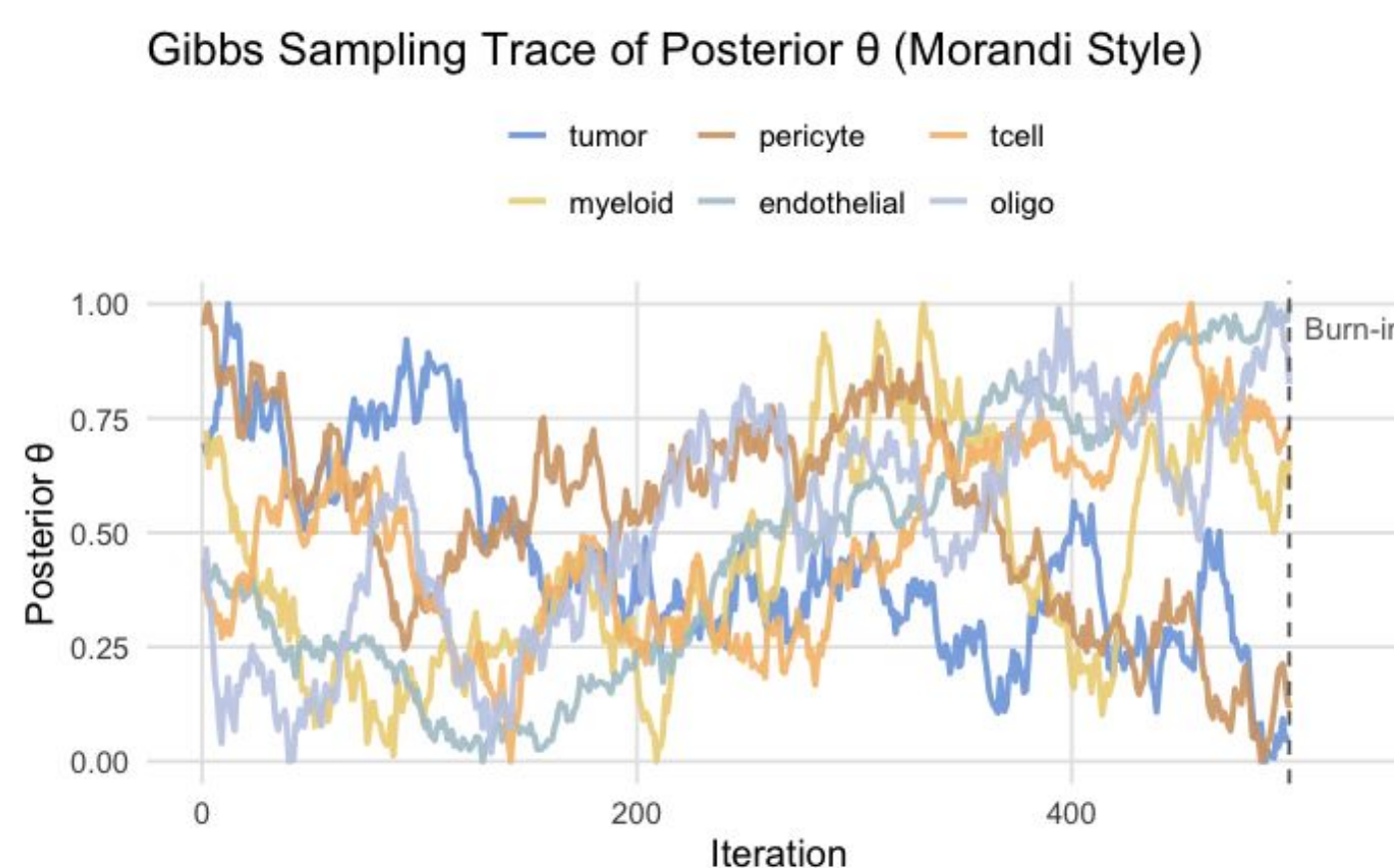


Bayesian Deconvolution of Bulk Transcriptomes.

BayesPrism We inferred heterogeneous compositions of tumor, myeloid, and pericyte cells from bulk RNA-seq data. Correlation analysis revealed cooperative and antagonistic relationships among cell types, delineating tumor-myeloid-pericyte interactions. in glioblastoma.

Posterior convergence diagnostics.

Gibbs sampling traces of cell-type proportions (θ) show acceptable convergence for major cell types. However, posterior densities between burn-in and stable phases indicate partial convergence, suggesting that further tuning (e.g., longer chains or adaptive step sizes) is needed to improve mixing.



Weighted Gene Co-expression Network Analysis I

To identify clusters of co-expressed genes (modules) that may represent biologically meaningful pathways, we performed a weighted gene co-expression network analysis (WGCNA) on normalized RNA-seq data.

Framework of WGCNA

Data processing $\Rightarrow$ Outlier removal $\Rightarrow$ Scale-free network construction $\Rightarrow$ Module construction $\Rightarrow$ Feature extraction.

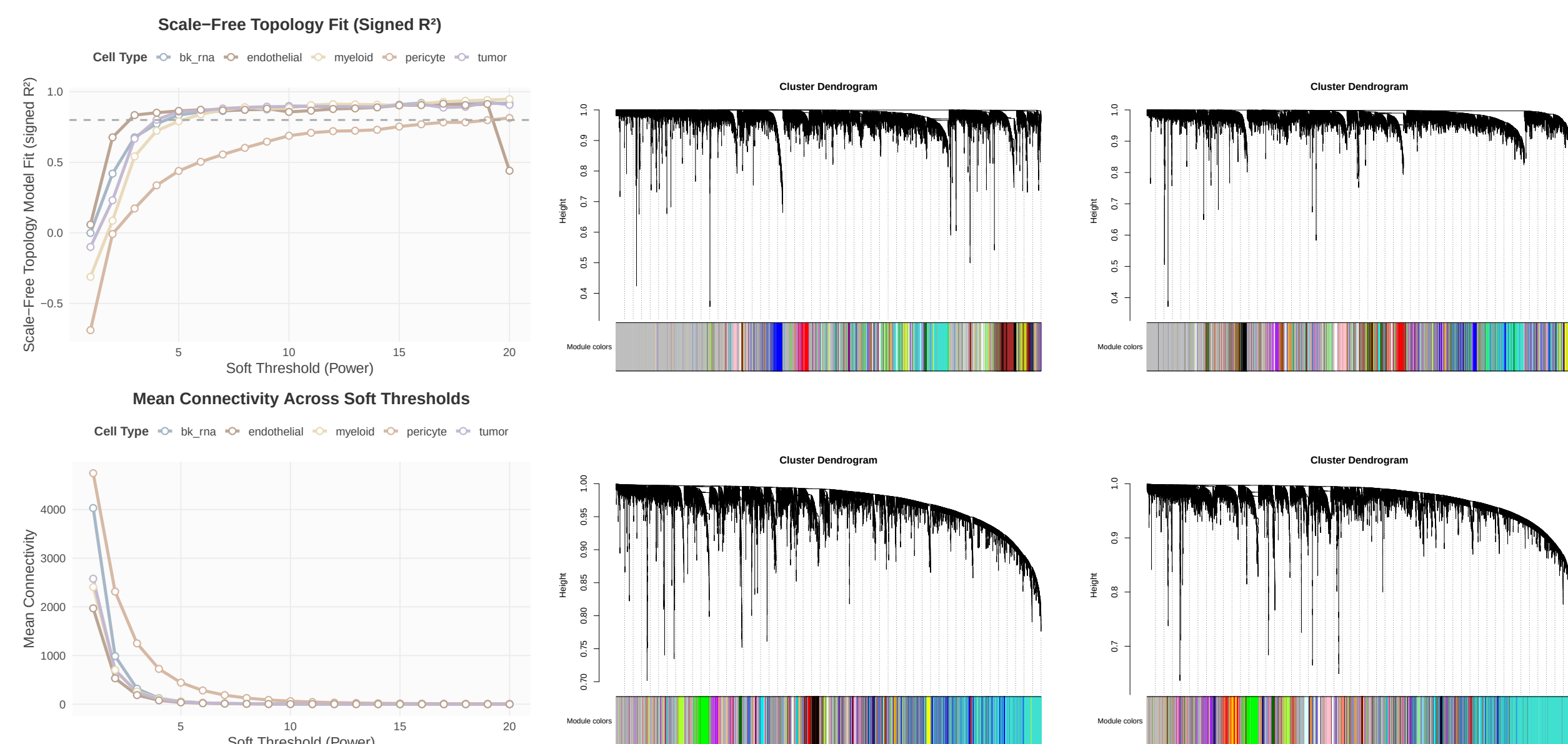
Data preprocessing and normalization. Genes expressed in fewer than 20% of samples (counts > 5) were removed. RNA-seq counts were normalized using the variance-stabilizing transformation in **DESeq2**, which corrects sequencing depth via sample-specific size factors s_j ($\hat{x}_{ij} = x_{ij}/s_j$). Variance stabilization is then applied: $y_{ij} = f(\hat{x}_{ij}) \approx \log_2(\frac{\hat{x}_{ij}}{\theta} + 1)$ where θ is a fitted dispersion parameter. This ensures: $y_{ij} = f(\hat{x}_{ij}) \approx \log_2(\frac{\hat{x}_{ij}}{\theta} + 1)$ allowing reliable downstream correlation computation.

Remove Outliers: Outlier samples were identified and removed using WGCNA's built-in function **goodSampleGenes()** and then hierarchical clustering using **hclust()**.

Scale-free network Construction: For normalized expression data y_{ij} , compute pairwise Pearson correlations. Then transform correlations into an adjacency matrix $A = (a_{mn})$ using a soft-thresholding power β : $a_{mn} = |r_{mn}|^\beta$. The parameter β was chosen as the spot that has the highest mean connectivity while maximizing an R^2 value, which usually occurs when $R^2 > 0.8$ in scale-free topology fit plot.

Weighted Gene Co-expression Network Analysis II

Module Construction: The **blockwiseModules()** function was used to create an unsigned adjacency matrix, transform it into an unsigned topological overlap matrix (TOM), and perform hierarchical clustering with the minimum module size of 20 and **mergeCutHeight** = 0.15. $TOM_{mn} = \frac{\sum_u a_{mu}a_{un} + a_{mn}}{\min(k_m, k_n) + 1 - a_{mn}}$ where $k_m = \sum_u a_{mu}$ is the connectivity of gene m .



Module feature extraction: Computed eigengenes (1st principal component) for each module.

Hub Gene

Hub Genes = High Centrality & Membership.

For each gene g , define:

$k_{\text{Within},g}$ = TOM-based connectivity,
 $kME_g = \text{corr}(x_g, \text{module eigengene})$.

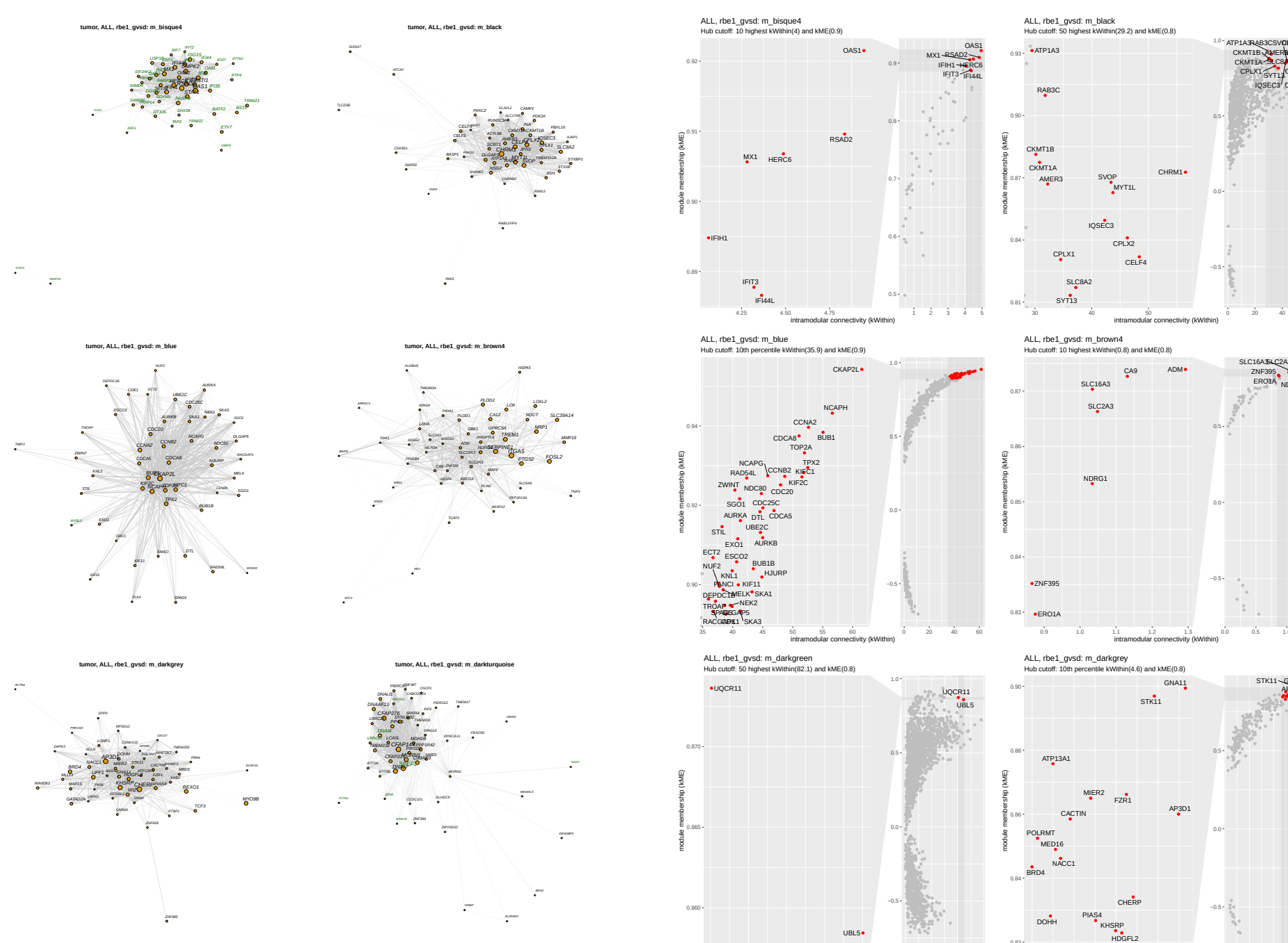
Under a latent factor model, both metrics increase with the factor loading $|\lambda_g|$:

$$k_{\text{Within},g}, kME_g \uparrow |\lambda_g|.$$

Let the set of hub genes be

$$\mathcal{H} = \{g : k_{\text{Within},g} > 0.8, kME_g > 0.8\}.$$

Genes above both thresholds are highly connected and representative, forming the structural cores of each co-expression module. Scatter plots show genes per module (x: $k_{\text{Within},g}$; y: kME_g); top-right points (red) denote hub genes, grey points denote others.



Enrichment Analysis

Functional Enrichment of Modules. Each module's gene set was analyzed with **Enrichr** to test overrepresentation in GO, KEGG, and WikiPathways. Enrichment is quantified by Fisher's exact test on a 2×2 table:

	In pathway	Not in pathway
In module	a	b
Not in module	c	d

$$p = \sum_{k=a}^{\min(m,K)} \frac{\binom{K}{k} \binom{N-K}{m-k}}{\binom{N}{m}}.$$

p -values are FDR-adjusted (Benjamini–Hochberg). Significant enrichment ($p_{\text{adj}} < 0.05$) indicates functional coherence. Heatmaps show top enriched terms (columns) across WGCNA modules (rows), with color intensity proportional to fold enrichment (odds ratio).

Survival Analysis

Objective. Identify eigengene modules (ME_k) and clinical traits predictive of survival in glioblastoma (GBM).

Survival Analysis Pipeline Overview.

$$\text{WGCNA} \xrightarrow{\text{module eigengenes}} \text{Cox (univariate)} \xrightarrow{p < 0.01} \text{Multivariate Cox / LASSO-Cox} \rightarrow \text{KM curves \& AUROC}$$

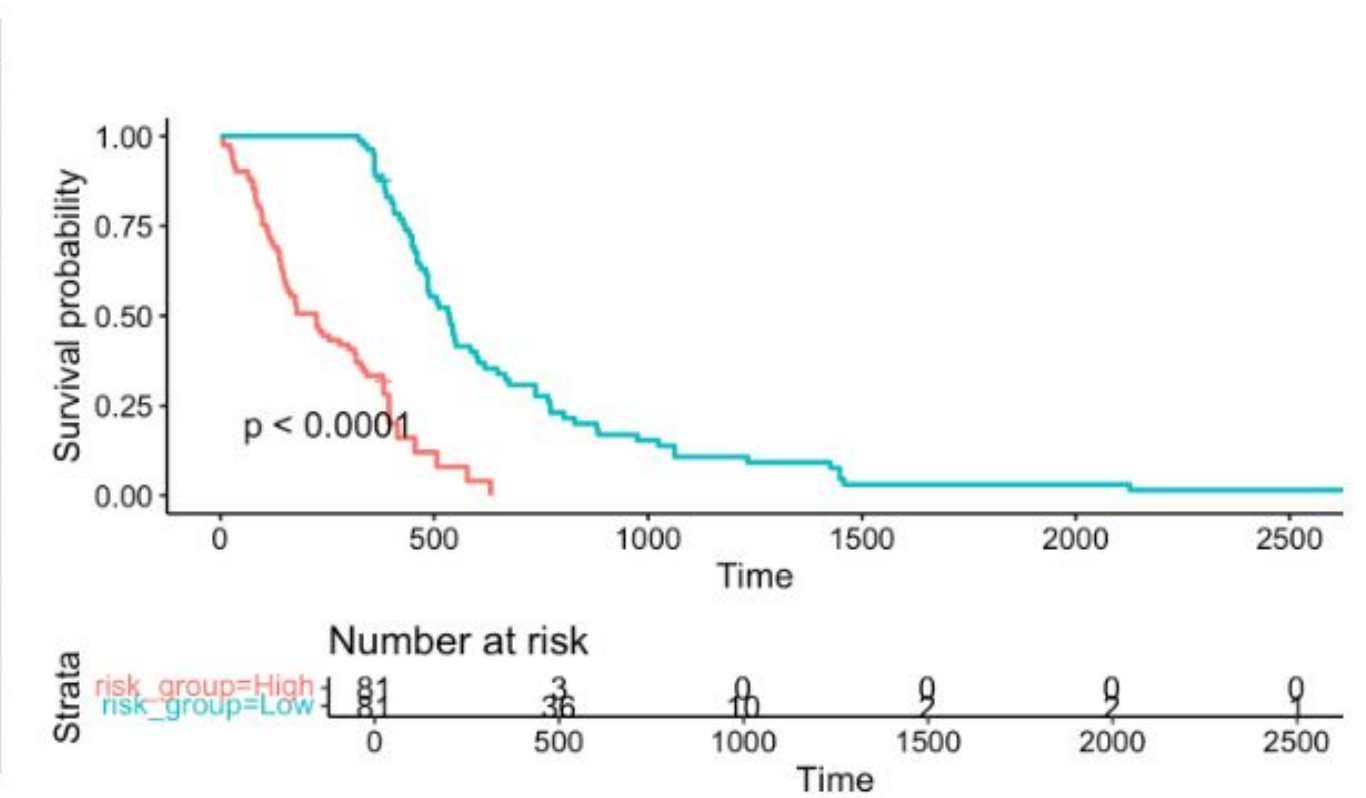
Model and Estimation.

$$\hat{\beta} = \arg \min_{\beta} \left\{ -\ell(\beta) + \lambda(\alpha \|\beta\|_1 + (1 - \alpha) \frac{1}{2} \|\beta\|_2^2) \right\}, \quad h(t|\mathbf{x}) = h_0(t) \exp(\beta_1 \text{Age} + \beta_2 \text{Treatment} + \sum_{k=1}^K \beta_k ME_k)$$

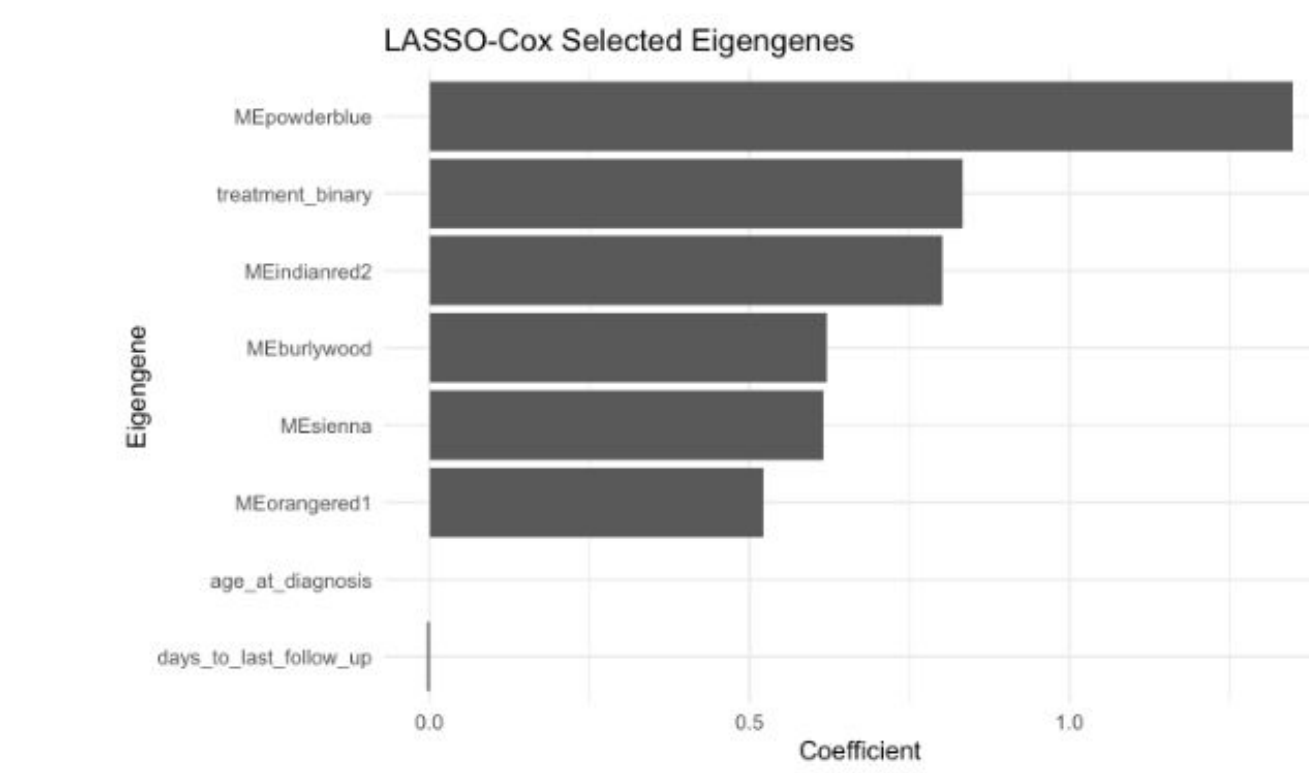
where $h_0(t)$ is the baseline hazard and β are estimated via LASSO ($\alpha = 1$) or Elastic Net ($\alpha = 0.5$).

Conclusion. LASSO-Cox identified $ME_{\text{powderblue}}$, $ME_{\text{indianred2}}$, and ME_{sienna} as key risk-associated modules ($\hat{\beta} > 0$), where higher expression predicts poorer survival. Regularization reduced collinearity and enhanced interpretability, highlighting these eigengenes as potential GBM survival markers. Future work will refine censoring logic, perform cross-validation for model stability, and validate findings in the TCGA GBM cohort.

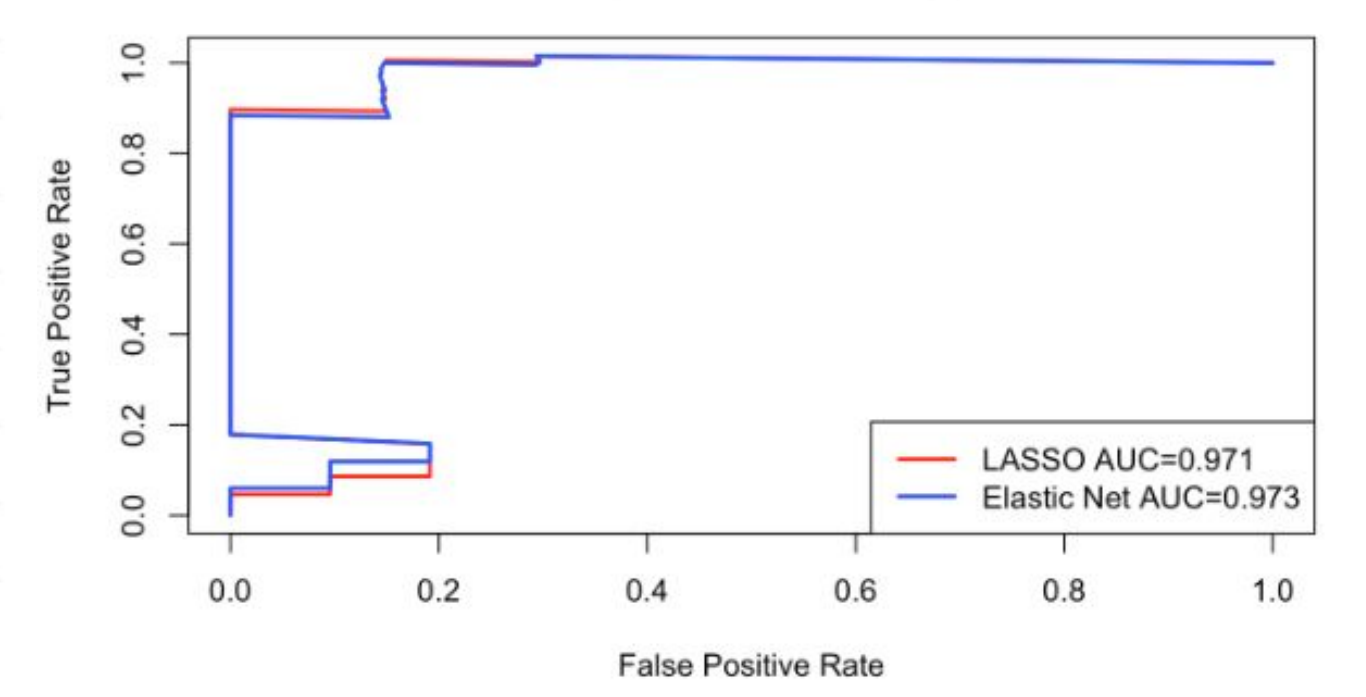
Variable	N	Hazard ratio	p
MEblue2	162	1.45 (0.00, 5930.87)	0.931
MEblueviolet	162	2.22 (0.10, 49.77)	0.616
MEbrown2	162	31.63 (0.61, 1643.92)	0.087
MEburlywood	162	2.64 (0.23, 31.01)	0.439
MEindianred2	162	0.35 (0.00, 142.14)	0.734
MEindianred3	162	0.00 (0.00, 0.42)	0.027
MEorange4	162	49.78 (0.00, 5329919.22)	0.508
MEorangered1	162	7.98 (0.08, 813.90)	0.379
MEpink3	162	4316.54 (2.28, 8158710.06)	0.030
MEpowderblue	162	38.97 (2.63, 577.36)	0.008
MEsalmon2	162	0.02 (0.00, 17.88)	0.257
MEsienna	162	47.74 (2.42, 941.04)	0.011
MEsienna3	162	14.20 (0.49, 410.37)	0.122
MEstateblue	162	1.91 (0.16, 22.57)	0.607
age_at_index	162	0.00 (0.00, 0.01)	0.003
age_at_diagnosis	162	1.00 (1.00, 1.00)	0.003
days_to_last_follow_up	162	1.00 (0.99, 1.00)	<0.001
treatment_binary	162	3.33 (1.85, 5.99)	<0.001



3. Lasso-Cox Selected Eigengenes Bar Plot of Coefficients

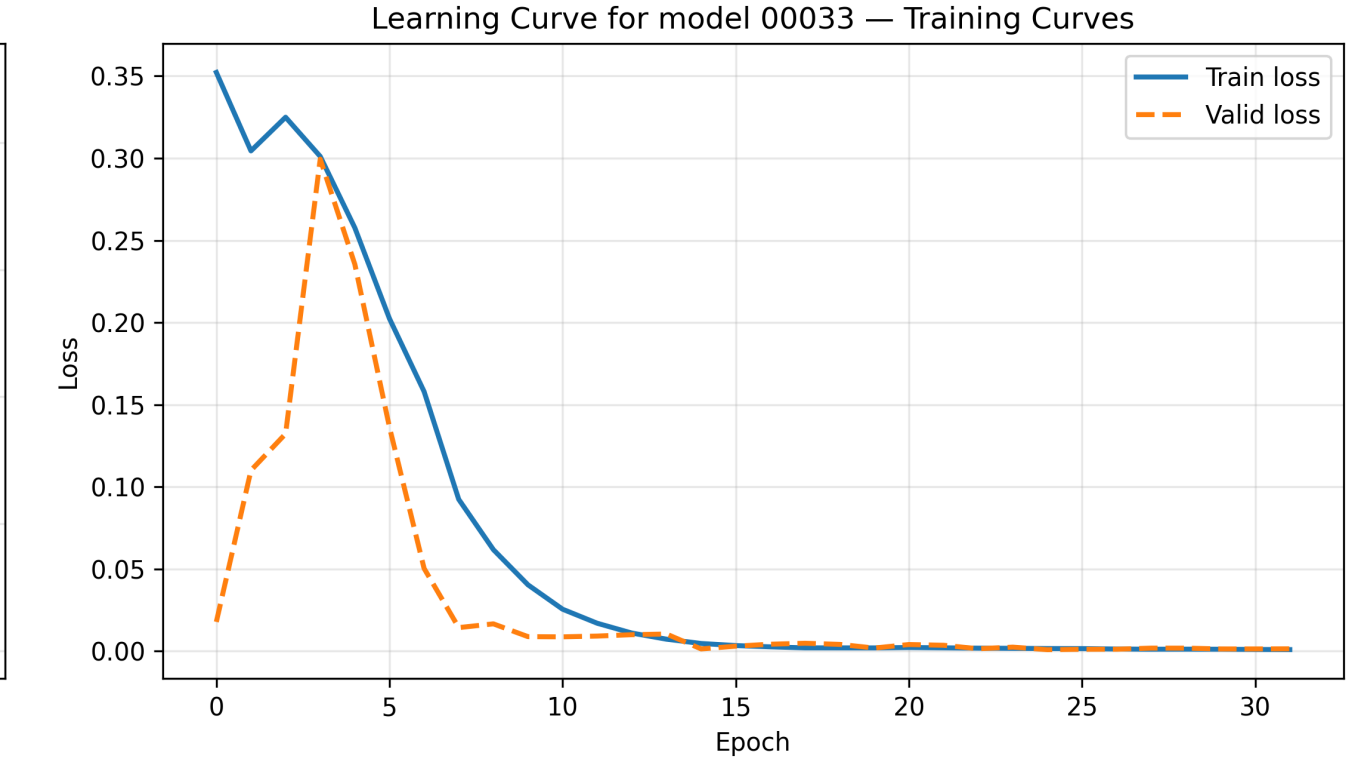
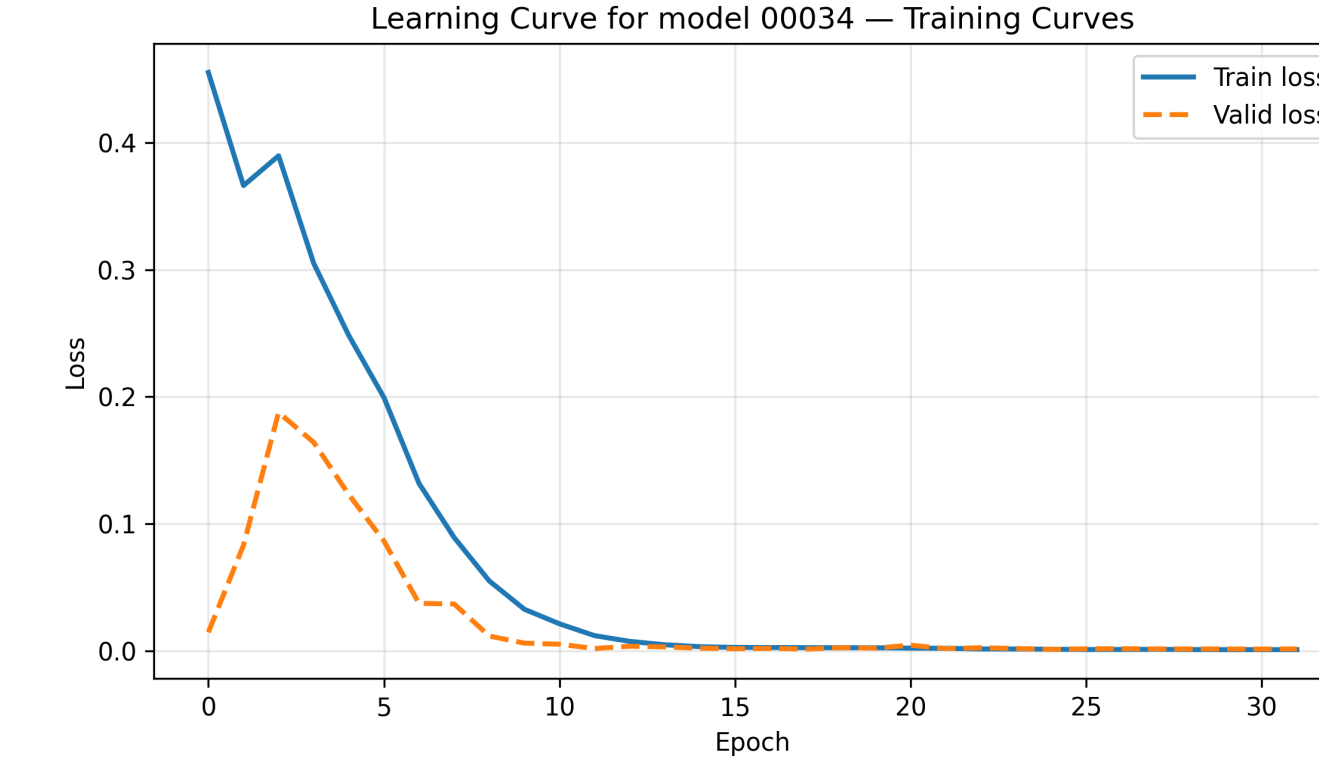
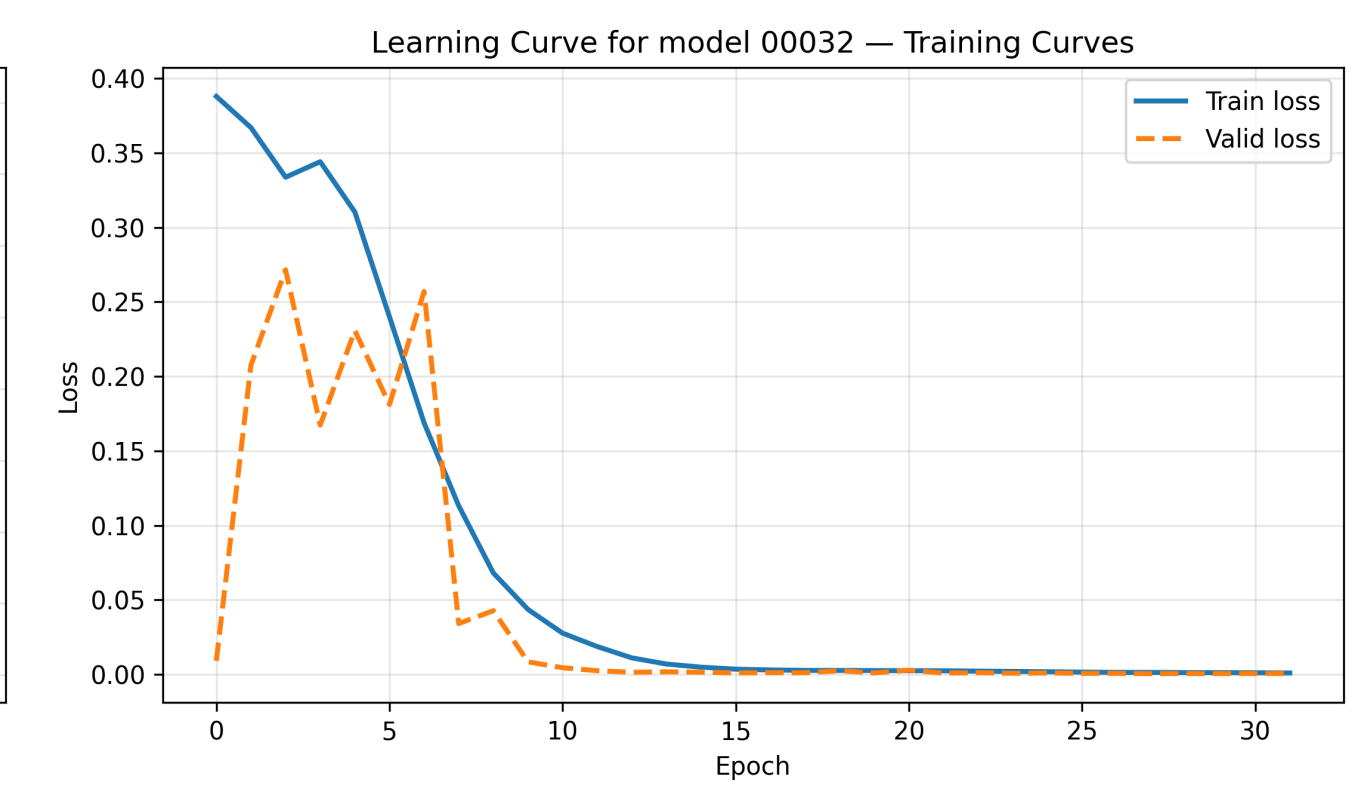
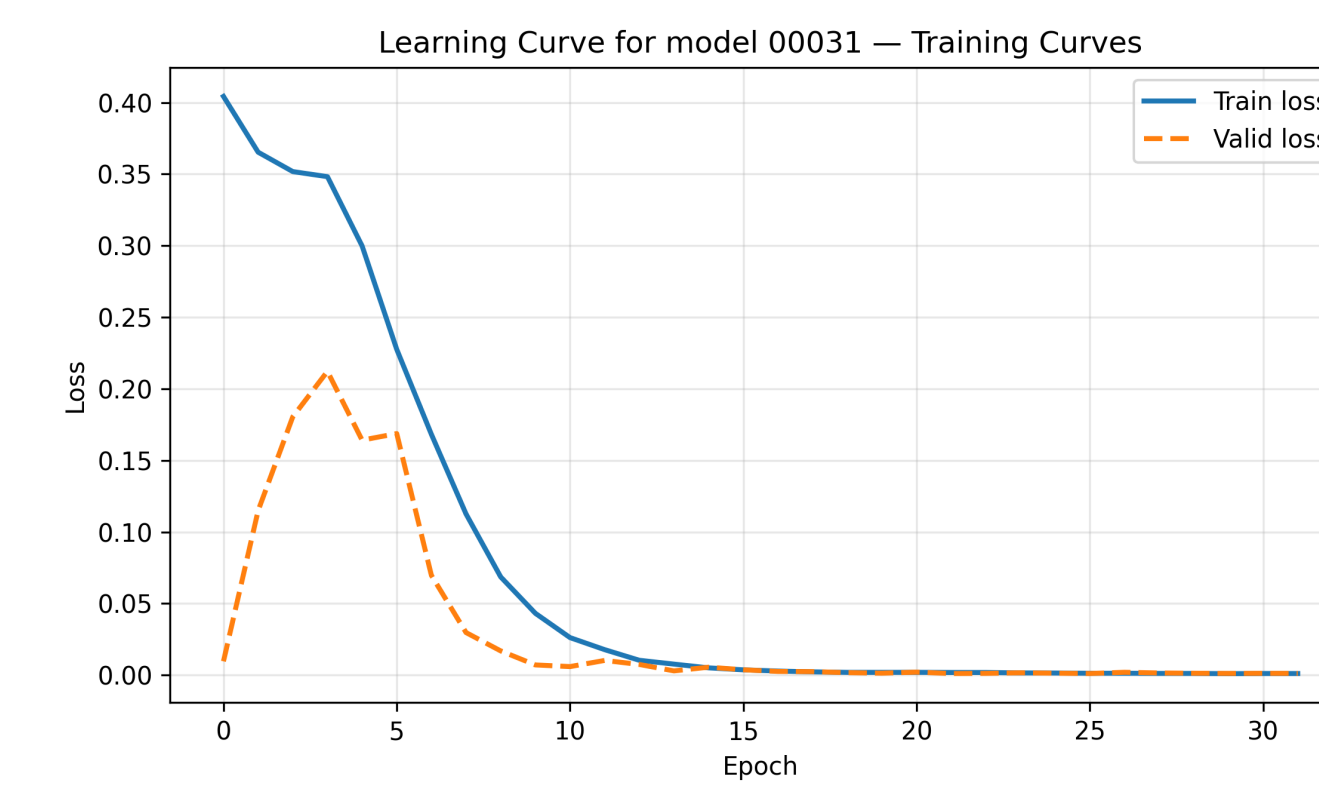


Time-dependent ROC at 3 years



SlideFlow

We used tumor WGCNA eigengene modules as targets and trained a multiple instance learning (MIL) model to predict the eigengene values directly from whole-slide pathology images. We're currently training these models and have already seen promising performance in the training phase



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